02_23_2021_copy

Tuesday, February 23, 2021 12:11 AM

Announcements

Project 1 - Due Friday @ 11:59 PM

Last Time

Miss Rate Reduction → Tiling
→ Loop interchange
→ profiling based I-cache re-arrangement

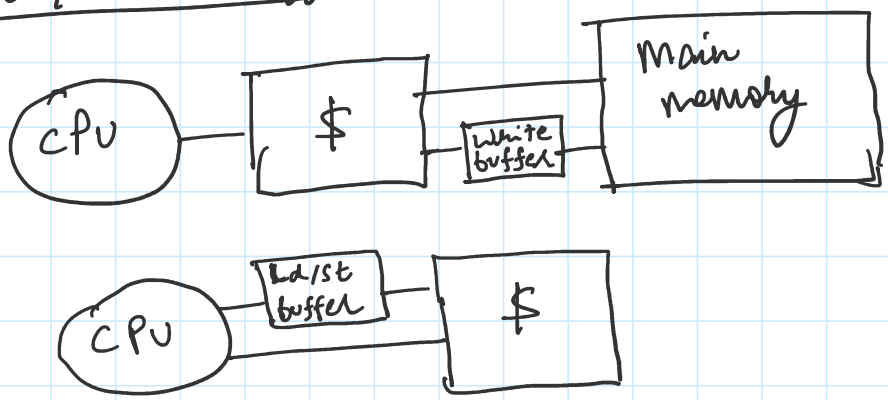
Today

Hit Time Reduction

Direct Mapped Caches
- No need to search members of the set
- Might need larger $ for equal miss rate
- sometimes page size doesn't let you do it

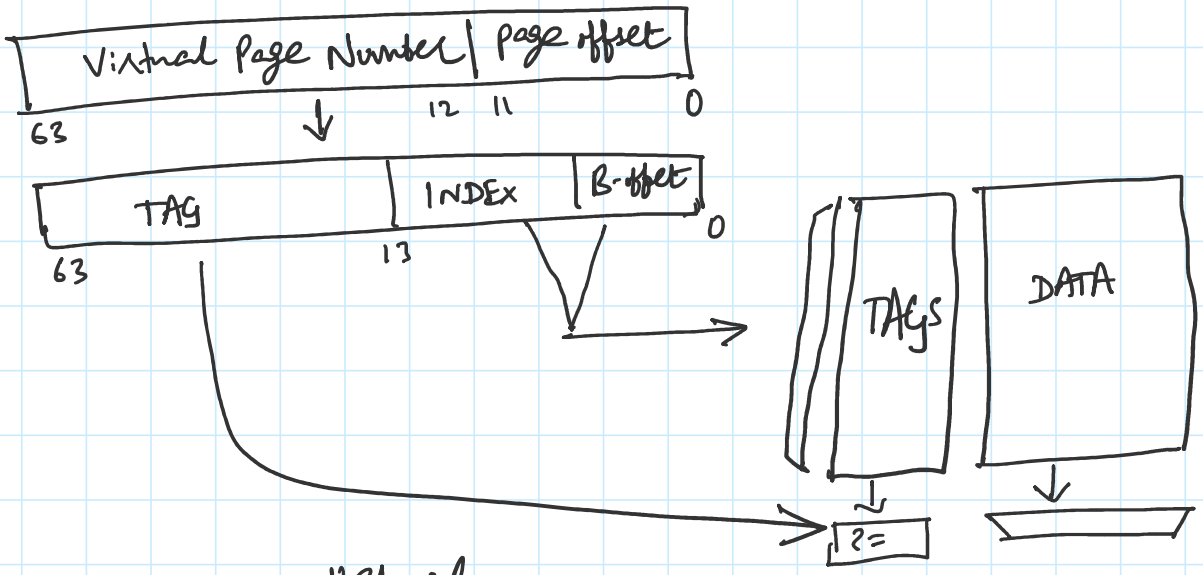
Small caches

Load / store Buffers



Accessed before the $ instead of at miss time
- writes sent to Ld/St buffer until flushed to $
- following reads can access Ld/St buffer

Virtually Indexed Virtually Tagged Caches

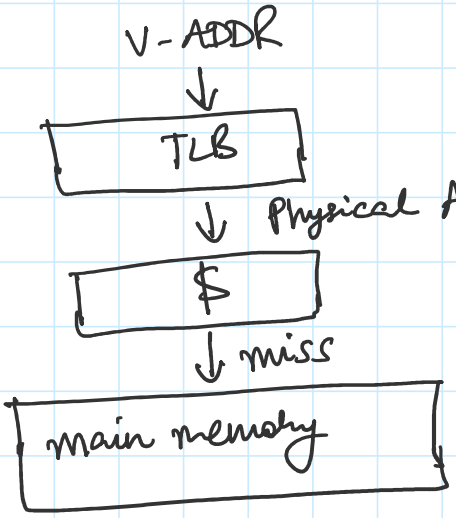


CPU Addresses - Virtual

- 2 processes can have same virtual Address
- shared memory between process same physical frame for different virtual addresses
- Every write will need to check a reverse translation Buffer and write to all blocks that have the same page frame

Virtually vs. Physically Addressed $

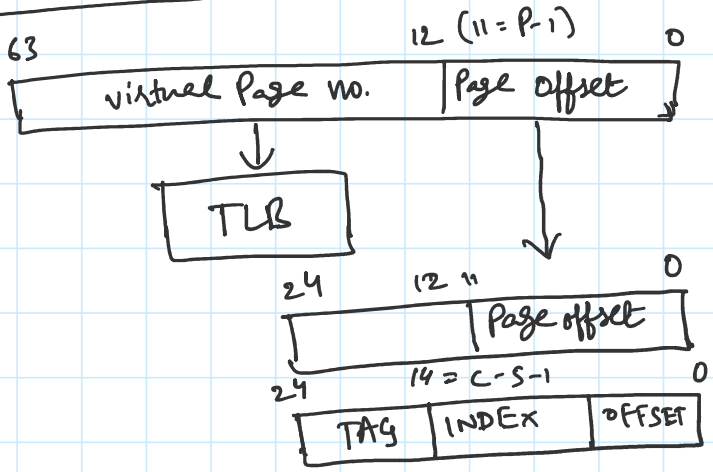
- special $, TLB (translation lookaside buffer) is checked for virtual address to physical address mapping
- On miss, $ fetches from physical memory



Problem!

Hit time ↑

Virtual Address Translation Details



Example

page size = 4kB

$P = \log_2(4kB) = 12$

32MB physical Memory

25-bits physical Address

Observation: What if index bits were entirely contained in page offset bits?

→ Can Access tag Array in II with TLB